Imperial College London

Lecture 10

Delay in Digital VLSI Circuits

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Key delay parameters

Propagation delay, t_{pd}

maximum time from the input crossing 50% to the output crossing 50%

Rise time, t_r

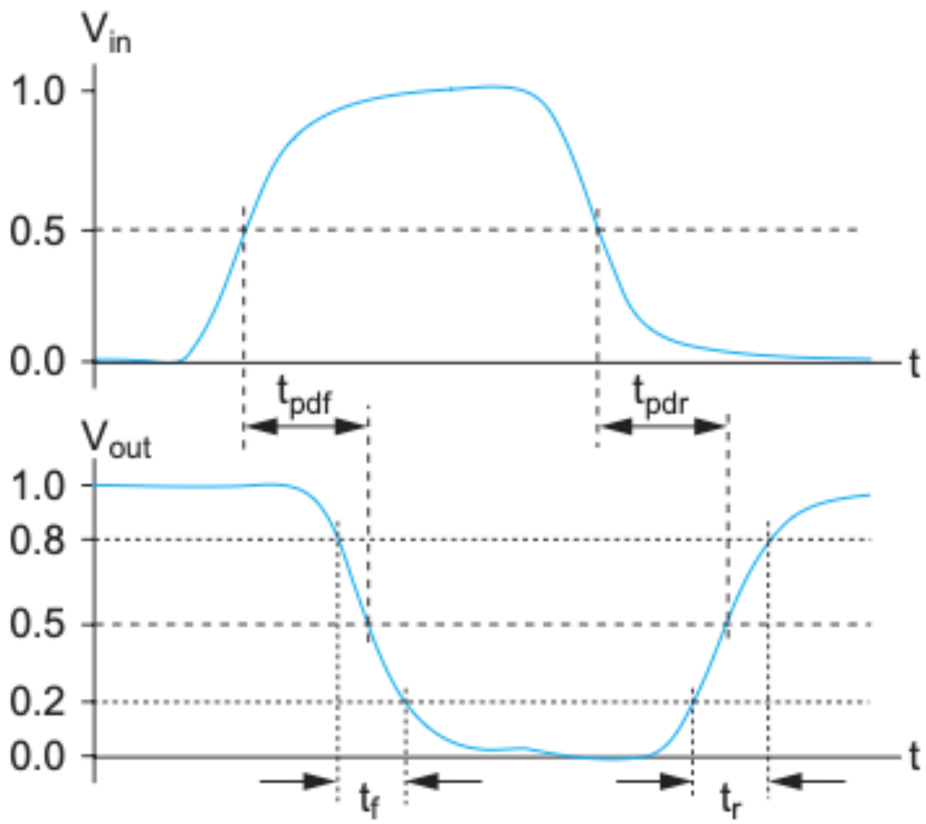
time for a waveform to rise from 20% to 80% of its steady-state value

Fall time, t_f

time for a waveform to fall from 80% to 20% of its steady-state value

Edge rate, t_{rf}

Average of rise and fall times, $(t_r + t_f)/2$



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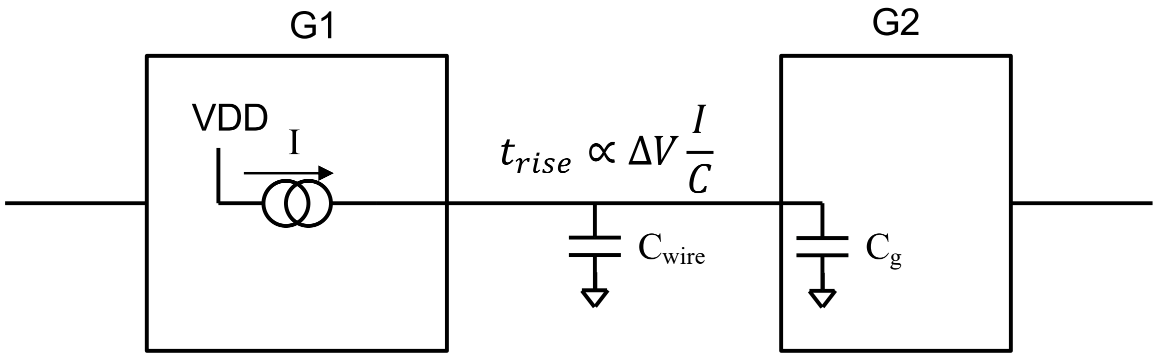
What causes delay in digital CMOS?

- **Nodes** - the wires between gates
- **Driver** - the gate that charges and discharges a node
- **Delay** – worst case propagation dela

- Delay mostly caused by two factors:
  1. Gate capacitance C_g
  2. Interconnection capacitance C_{wire}

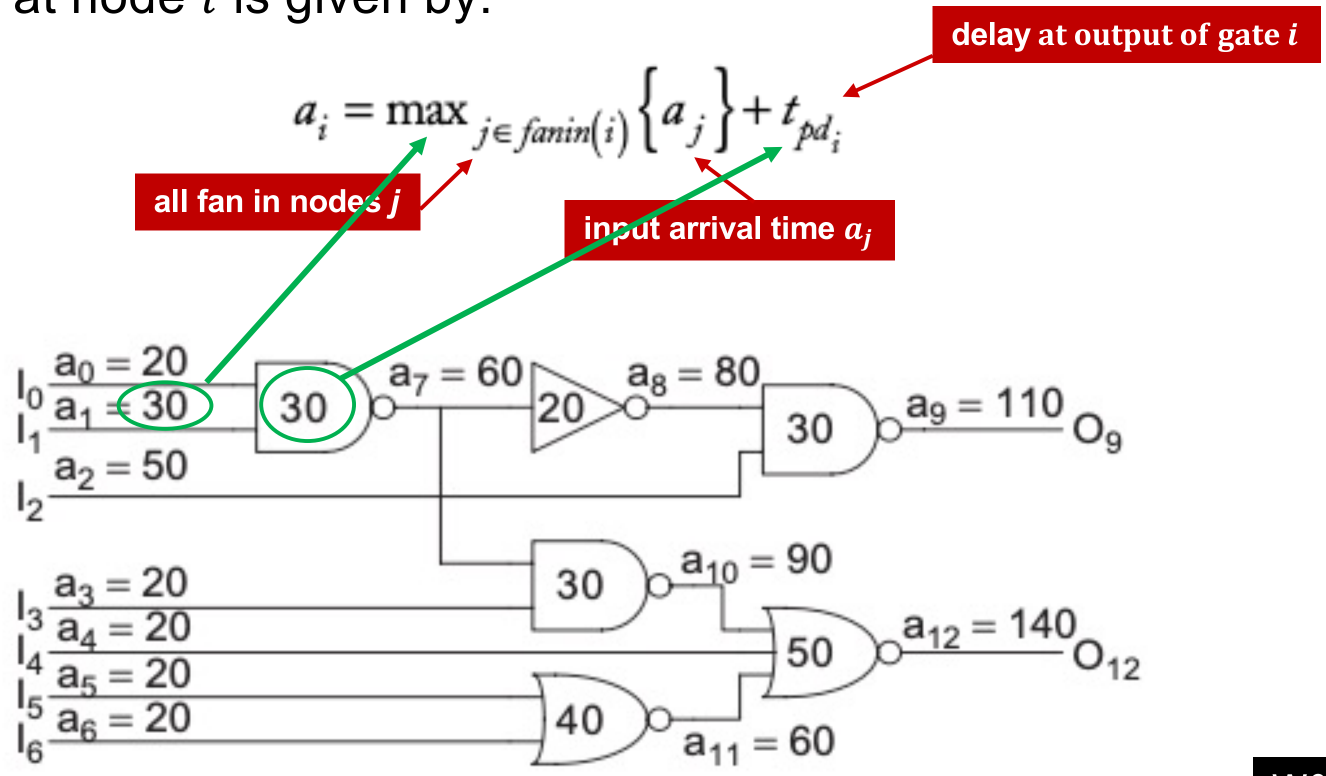
- Propagation delay of G1 $t_{pd} \approx$ time to charge/discharge its output



Arrival time

- Timing Analyzer computes arrival times in circuit nodes in module.
- User specifies arrival time of signals at inputs.
- User specifies required arrival time at outputs.
- Arrival time a_i at node i is given by:

$$a_i = \max_{j \in fanin(i)} \{a_j\} + t_{pd_i}$$



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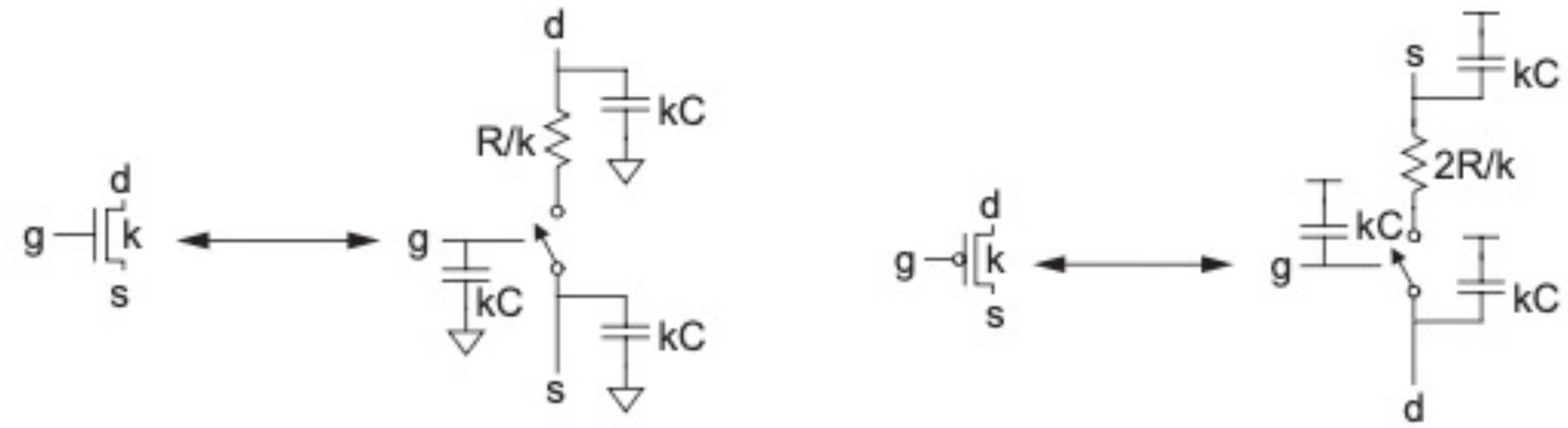
RC model of nMOS and pMOS transistors

- A unit transistor is an nMOS transistor in a minimum size inverter.
- R is the resistance, and C is gate capacitance of a unit transistor.
- nMOS transistor k x width of unit transistor has channel resistance = R/k.
- Its gate, drain and source capacitance is kC.
- pMOS mobility is around ½ that of nMOS, therefore, to have equal rise and fall times, pMOS transistor is twice as wide as nMOS transistor.
- Hence the equivalent delay model for nMOS and pMOS transistors are:



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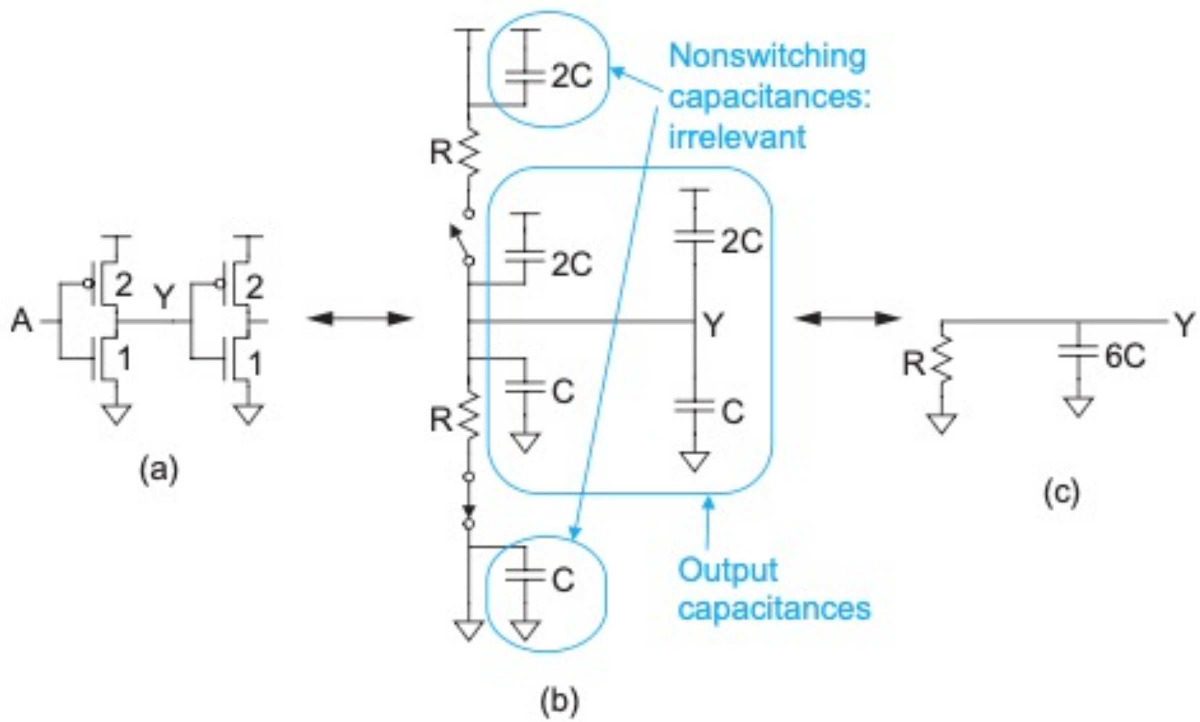
RC model of an inverter

- Inverter with fanout of 1 (driving one unit inverter).
- Inverter has unit size (i.e. minimum).
- Input $0 \rightarrow 1$, the equivalent circuit is shown in b).
- Therefore, the RC model is output node driven by R and 6C.



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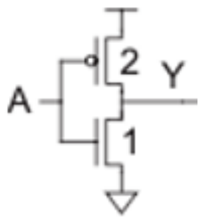
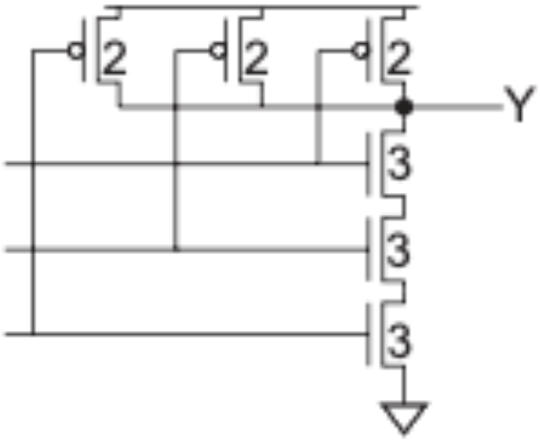
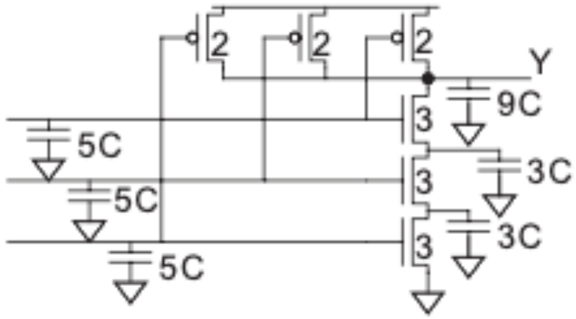
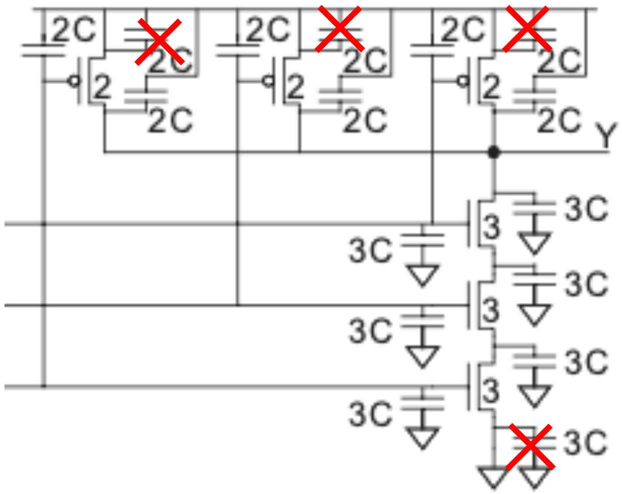
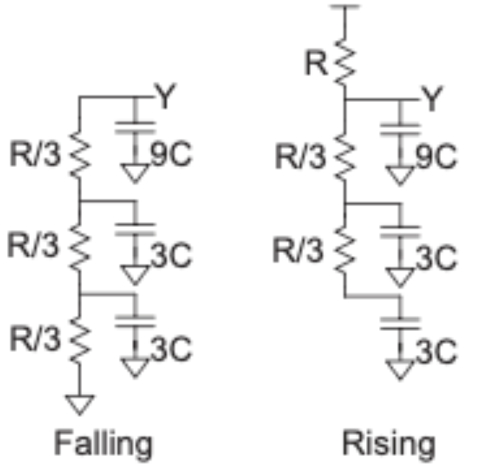
RC model of a 3-input NAND gate











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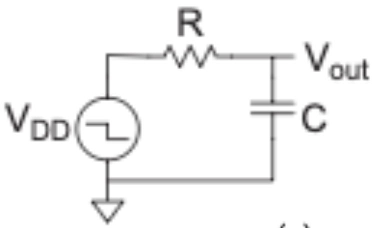
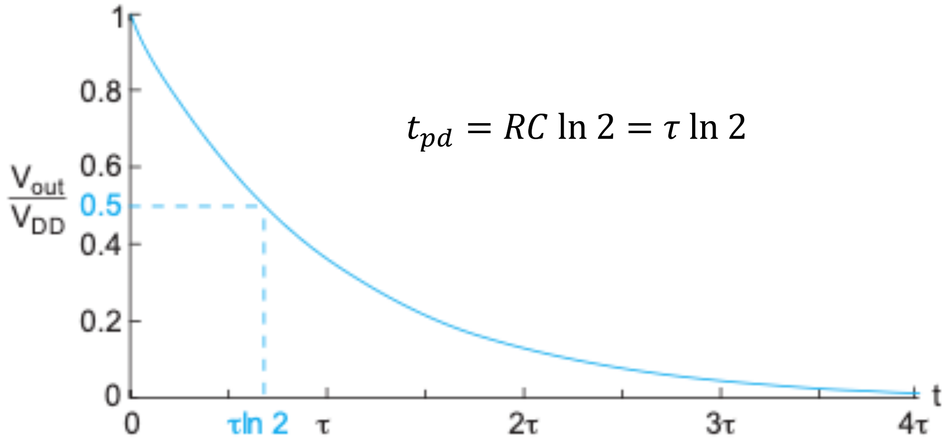
Gate delay estimation using RC model

1st order model

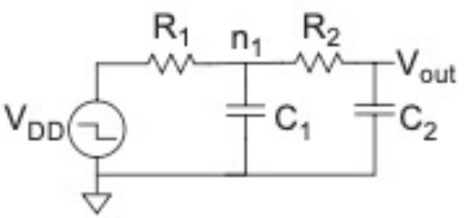


$$V_{out}(t) = V_{DD} e^{-t/\tau}$$

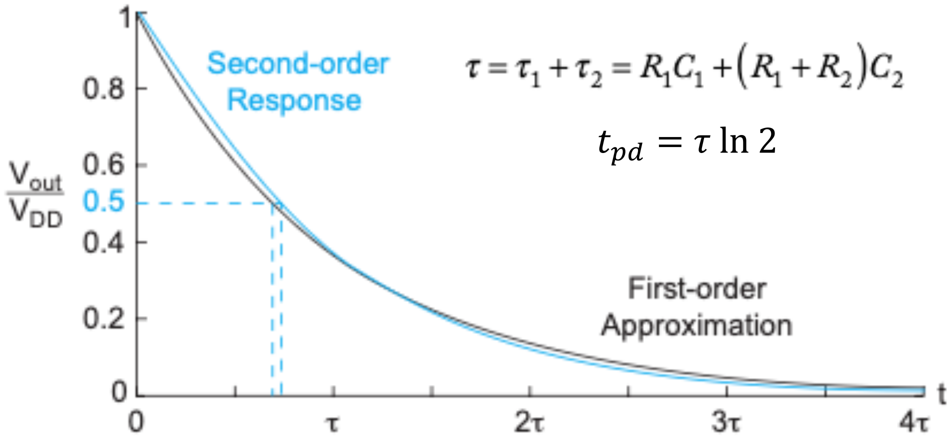
$$H(s) = \frac{1}{1 + sRC}$$



2nd order model



$$H(s) = \frac{1}{1 + s\left[R_1C_1 + (R_1 + R_2)C_2\right] + s^2 R_1C_1R_2C_2}$$



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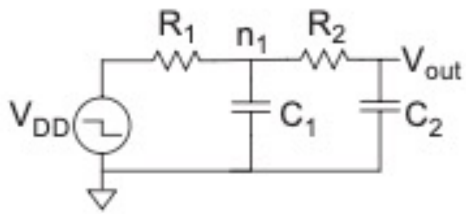
Elmore Delay Model for RC tree

- Real circuit has many stages of RC (tree of RC circuits).
- Elmore delay model estimate delay at node i *to be*:

$$t_{pd} = \sum_i R_{is} C_i$$

1. From source VDD to Vout, there is a leave node n1 and output node Vout.
2. For each node, the delay is the time constant given by the resistance from source (VDD) to node, multiply by node capacitance. For n1, $t_{pd_{n1}} = R_1 C_1$. For Vout, $t_{pd_{V_{out}}} = (R_1 + R_2) C_2$.
3. Total delay is the sum of these two.



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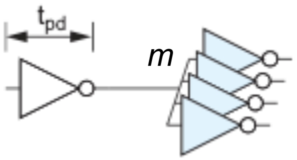
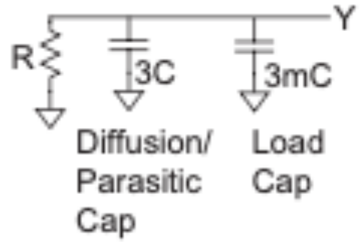
Examples of using Elmore Delay Model

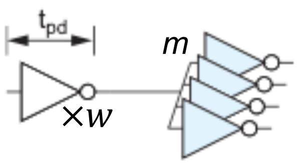
- Estimate t_{pd} for a unit inverter driving *m* identical unit inverters.



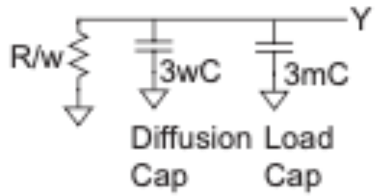
  - Total inverter input cap = $3mC$
  - Parasitic output capacitance = $3C$
  - $t_{pd} = 3(m+1)RC$



- Do the same, but now the driver inverter transistor with is w times unit size.



  - Total inverter input cap = $3mC$
  - Parasitic output capacitance = $3wC$
  - Effective resistance = R/w
  - Define fanout $h = load\ cap\ \div\ input\ cap = \frac{m}{w}$
  - $t_{pd} = \left((3w+3m)C\right)\frac{R}{W} = (3+3\text{h})RC$



- If a unit transistor has R= 10 kΩ and C = 0.1 fF, compute the delay, in picoseconds, of the inverter driving four identical inverter.
  - $RC = 10k\Omega \times 0.1fF = 1ps$
  - $h = 4,$ hence $t_{pd} = (3 + 3\times4)\times1ps = 15ps.$

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Linear Delay Model

- RC model → linear function of gate's fanout.
- Simplify delay analysis further using normalized delay in units of τ.
- $\tau \approx 3ps$ for 65nm process .
- Linear model:

delay $\boldsymbol{d}$ = gate's effort delay $\boldsymbol{f}$ + parasitic delay $\boldsymbol{p}$

gate's delay effort $\boldsymbol{f}$ = logical effort $\boldsymbol{g}$ × fanout $\boldsymbol{h}$

- The complexity of the delay model is represented by the **logical effort** $\boldsymbol{g}$
- An inverter (simplest gate) is defined to have a logic effort $\boldsymbol{g = 1}$.
- More complex gate has higher logic effort. For example, a 3-input NAND gate has an effort of 5/3.
- The fanout $\boldsymbol{h}$ is called the **electrical effort** $\boldsymbol{h}$, and is defined as:

$$h = \frac{C_{out}}{C_{in}}$$

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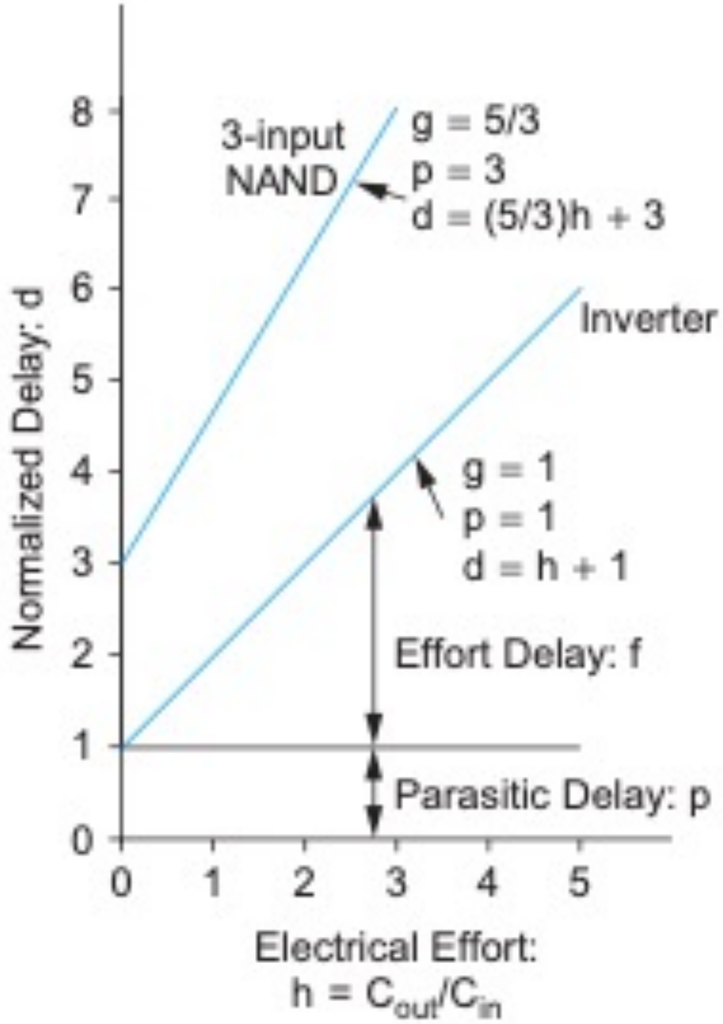
Delay vs Electrical Effort (fanout load)

- Model assumes linear relationship between delay and fanout load (electrical effort).
- For inverter, the plot has a gradient of 1.
- Inverter used as reference.
- The logical effort is the gradient (similar to gain of an amplifier).
- The intercept on the y-axis is the parasitic delay (no load).

- The upper plot is the same characteristic for a 3-input NAND gate.
- The logic effort g is 5/3, and that is the gradient of the line.
- The delay is:

$$d = g \times h + p = \left(\frac{5}{3}\right) h + 3.$$



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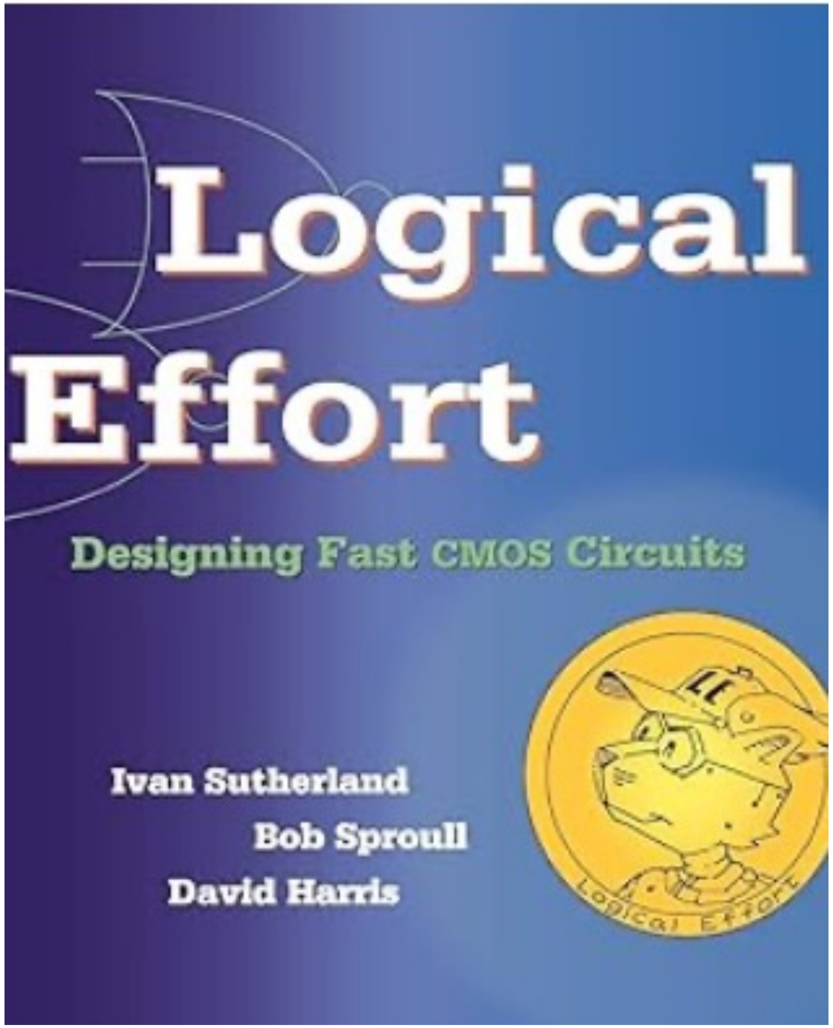
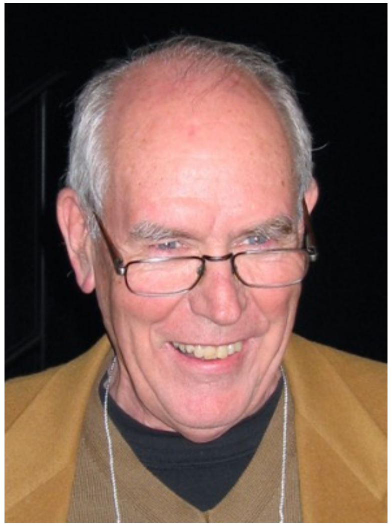
Method of Logic Effort



- Introduced by Ivan Sutherland during his sabbatical at Imperial College in early 1990's.
- A simple "back of envelop" calculation of delay.
- Optimize sizes of transistors for fast operation without using software tools.

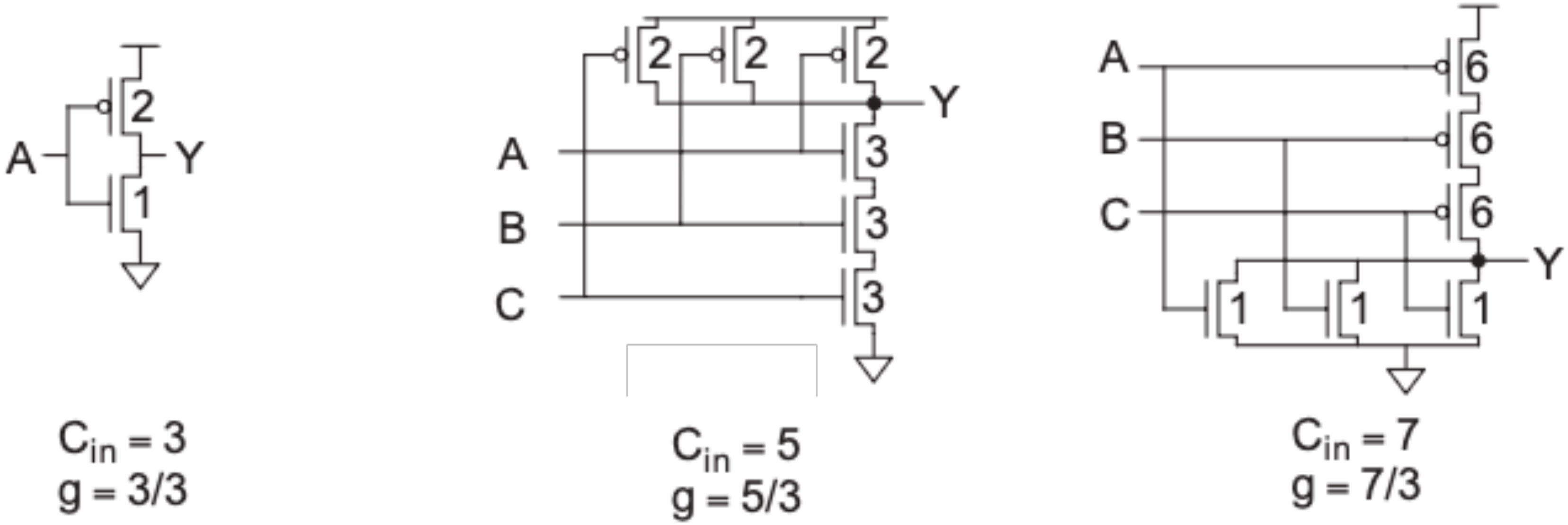


What is Logical Effort?

- Logical effort = input capacitance of the gate / input capacitance of an inverter that can deliver the same output current.

Logical Effort: $\boldsymbol{g} = C_{in_{gate}} / C_{in_{inv}}$

Effort delay $\boldsymbol{f}$ = logical effort $\boldsymbol{g}$ $\times$ electrical effort $\boldsymbol{h}$



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Logical Effort of Common Gates

Gate Type	Number of Inputs				
	1	2	3	4	n
inverter	1				
NAND		4/3	5/3	6/3	$(n+2)/3$
NOR		5/3	7/3	9/3	$(2n+1)/3$
tristate, multiplexer	2	2	2	2	2
XOR, XNOR		4, 4	6, 12, 6	8, 16, 16, 8	

- Logical effort increases with number of inputs.
- Logical effort for XOR/XNOR gates are different for different inputs.

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Parasitic Delay of Common Gates

Gate Type	Number of Inputs				
	1	2	3	4	n
inverter	1				
NAND		2	3	4	n
NOR		2	3	4	n
tristate, multiplexer	2	4	6	8	$2n$

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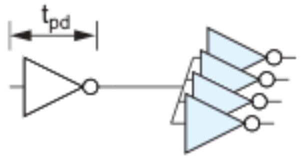
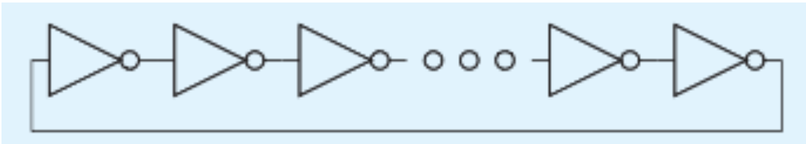
Estimate logic delay using Logical Effort

- Estimate the delay of a fanout-of-4 (FO4) inverter in L5S10.
  - Logical effort of inverter g = 1. Electrical effort h = 4.
  - Parasitic delay $p_{inv} \approx 1$.
  - Total delay $d = gh + p = 5$.



- A ring oscillator is constructed from an odd number of inverters. If a 31-stage ring oscillator has an output frequency of 2.7GHz, what is the unit delay of the inverter?



- Logical effort of inverter g = 1. Electrical effort h = 1.
- Parasitic delay $p_{inv} \approx 1$. Total delay for each inverter: $d = gh + p = 2$.
- N stage delay = 2N = half period of oscillation.
- $Period\ of\ oscillation = \frac{1}{2.7GHz} = 2 \times 2 \times 31 \times\ t_{pd_{inv}}$

$t_{pd_{inv}} = \tau = 3ps$

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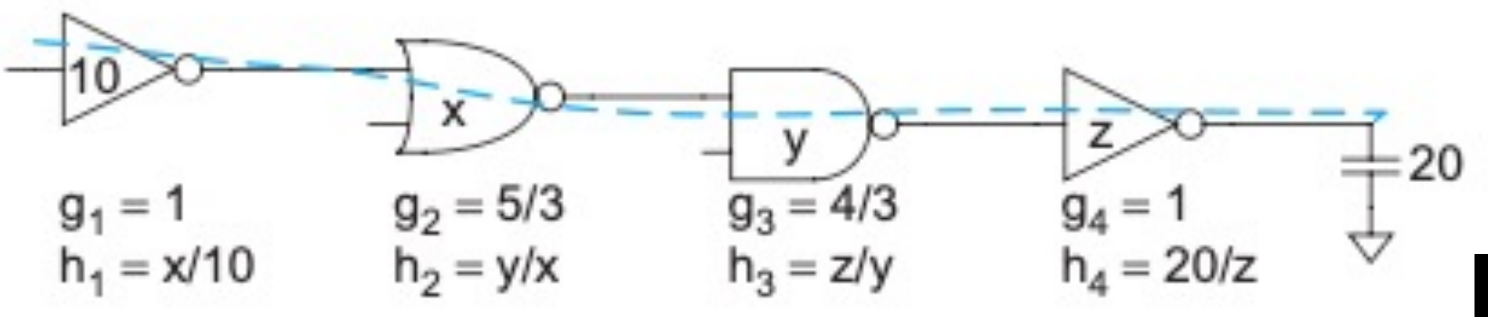
Apply Logical Effort to Multistage Network

- One application of logic effort is to size the transistor in a multistage network to optimize its delay.
- Logical effort can help to determine best sizes for x, y and z for driving a capacitance load of 20.
- Define Path logical effort G as: $G = \prod g_i$
- Define Path electrical effort H as: $H = \frac{C_{out(path)}}{C_{in(path)}}$
- Then, the Path effort is: $F = \prod f_i = \prod g_i h_i$
- Note that $F \neq GH$, unlike the case of individual stage where $f = gh$



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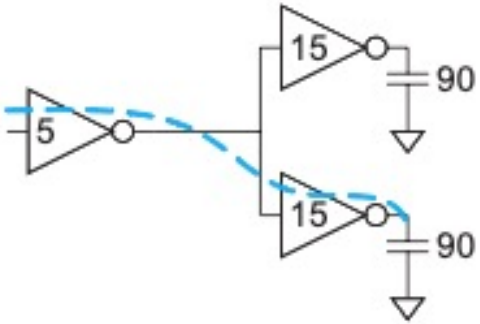
Branching Effort

- Only half the output current of the input inverter is used to drive the blue path. We therefore need something to account for this branching effect.
- **Branching effort *b*** of a branching node is: $b = \frac{C_{onpath} + C_{offpath}}{C_{onpath}}$.
- **Path branching effort *B*** is: $B = \prod b_i$
- Now we can define the **path effort *F*** as: $F = GBH$.



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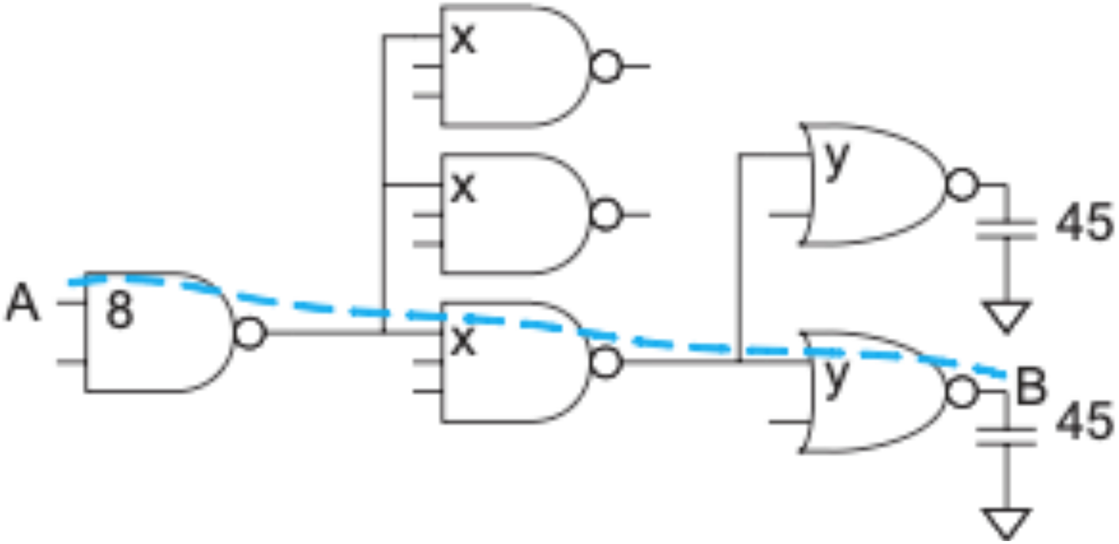
Example of using Logical Method (1)

Problem:

- Input NAND2 presents has capacitance of 8 transistor widths.
- Output load is equivalent to 45 transistor widths.
- Estimate the minimum delay of the path from A to B.
- Choose transistor sizes to achieve this delay.



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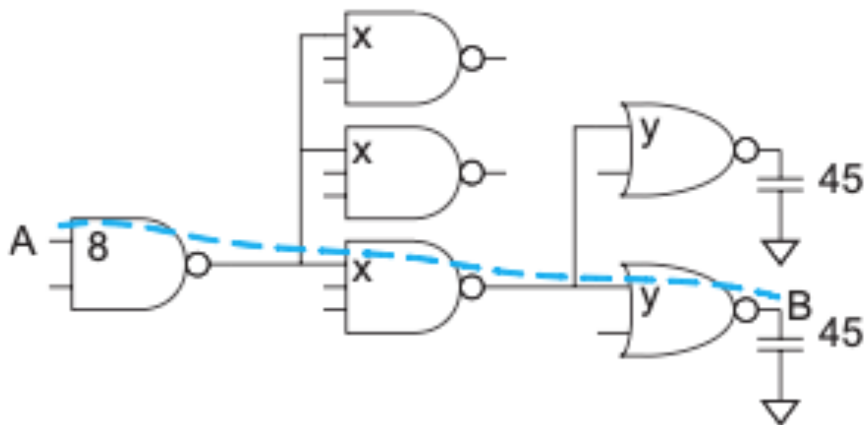
Example of using Logical Method (2)

Solution to best delay calcu;ation:

- Path logical effort G = (4/3) x (5/3) x (5/3) = 100/27.
- Path electrical effort H = 45/8.
- Path branching effort B = 3 x 2 = 6.
- Total path effort F = GBH = 125.
- Best stage effort = divide effort equally in 3 stage $\hat{f} = \sqrt[3]{125} = 5$.
- Path parasitic delay P = 2+3+2 = 7.
- Therefore, minimum path delay is D = 3 x 5 + 7 = 22 in units of τ.
- $\tau \approx 3ps$ for 65nm. Therefore, delay is 66ps!



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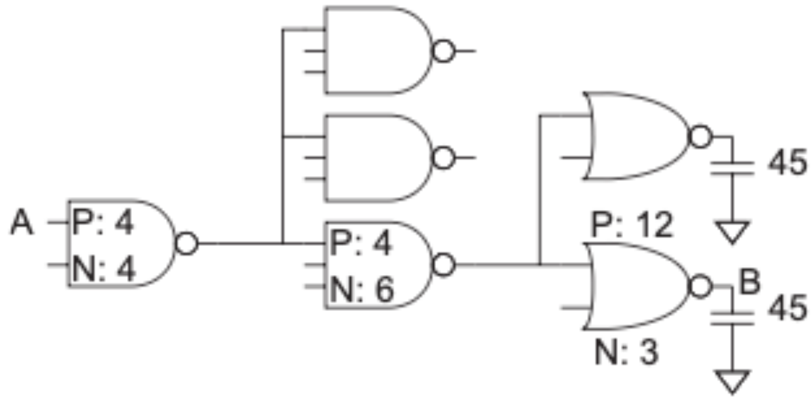
Example of using Logical Method (3)

Solution to sizes of transistors:

- Stage effort $\hat{f} = 5$.
- We know that $\hat{f} = g \times h = g \times \frac{C_{out}}{C_{in}}$ for each stage.
- Now, work from output backward:
  - NOR2 $C_{out} = 45$, therefore $C_{in} = \frac{g \times C_{out}}{\hat{f}} = \frac{\frac{5}{3} \times 45}{5} = 15$.
  - Therefor the NOR2 gate has pMOS width of 12, and nMOS width of 3.
  - NAND3 $C_{out} = 15 + 15 = 30$, therefore $C_{in} = \frac{g \times C_{out}}{\hat{f}} = \frac{\frac{5}{3} \times 30}{5} = 10$.
  - Therefore the NAND3 gate has pMOS width of 4, and nMOS width of 6.



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Choosing the Best Number of Stages

- For N-stage network with path effort F, minimum possible delay D is:

$$D = NF^{\frac{1}{N}} + P$$

where P is the path's parasitic delay.
- Key result from Logical Effort – no need to know sizes of transistors.
- How to choose N?
- Example of an inverter chain (see Example 4.14 in textbook).
- Path electrical effort $H = \frac{64}{1} = 64.$
- Path logical effort $G = 1.$
- Branching effort $B = 1.$
- Path effort $F = GBH = 65.$
- Best number of stage $N = \sqrt[N]{64}.$
- Delay $D = N\sqrt[N]{64} + N$

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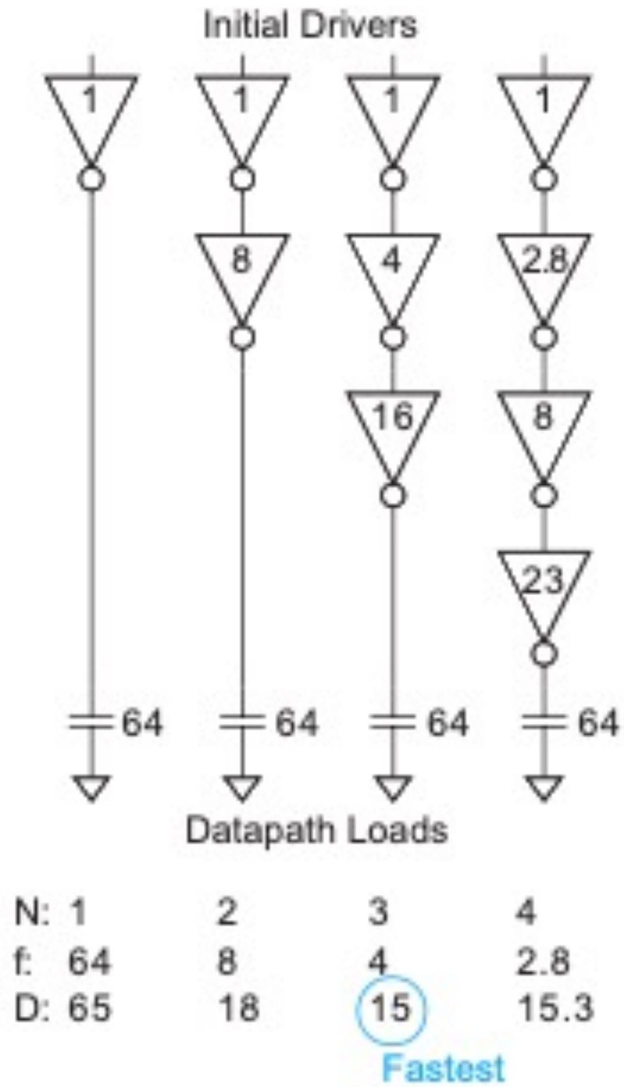


Adding inverter stages to optimize timing

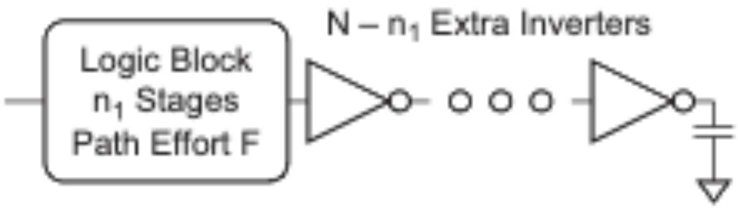
- Best number of stages = N
- If N know, can always add inverters (buffers).
- How to find $\hat{N}$ for a given path effort F?
- Theoretically shown by Mead and Conway that:

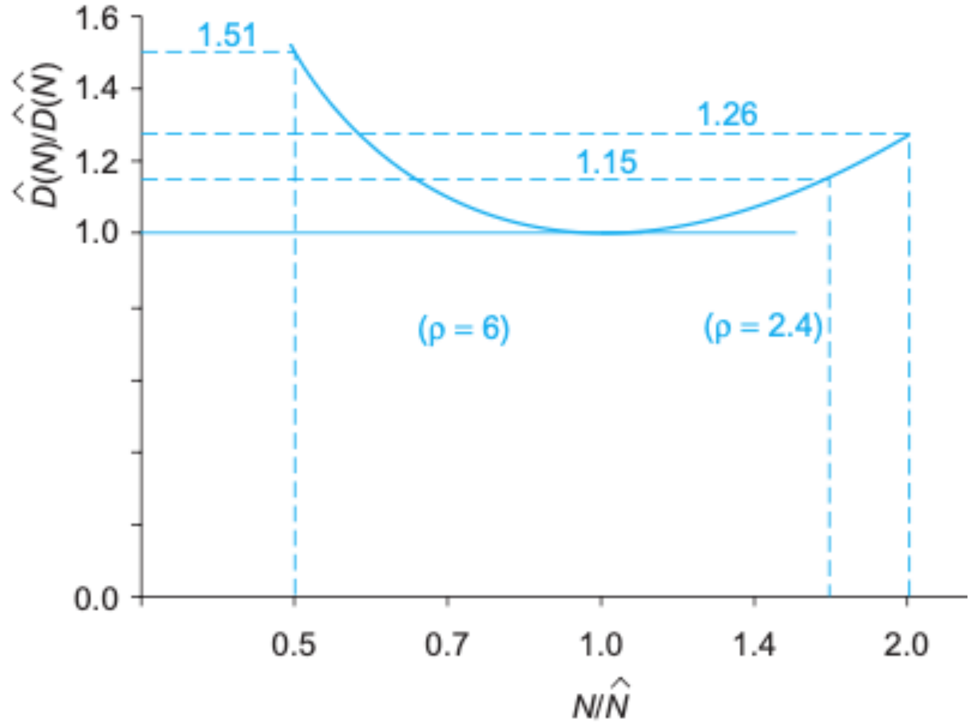
  $$\hat{N} = \log_e F, \text{ where e = 2.718.}$$

- Harris use simulation to show that, for chain of inverters, best number of stages is:

  $$\hat{N} = \log_4 F.$$

- The graph shows the sensitivity of delay to the number of stages





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Method of Logical Effect - Summary

1. Compute path effort. $F = GBH$
2. Estimate best number of stages. $N = \log_4 F$
3. Sketch path with N stages.
4. Estimate least delay. $D = NF^{\frac{1}{N}} + P$
5. Determine best stage effort. $\hat{f} = F^{\frac{1}{N}}$
6. Find gate sizes. $C_{in_i} = \frac{g_i C_{out_i}}{\hat{f}}$

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Logical Effort Equations

Term	Stage Expression	Path Expression
number of stages	1	N
logical effort	g (see Table 4.2)	$G = \prod g_i$
electrical effort	$h = \dfrac{C_{\text{out}}}{C_{\text{in}}}$	$H = \dfrac{C_{\text{out(path)}}}{C_{\text{in(path)}}}$
branching effort	$b = \dfrac{C_{\text{onpath}} + C_{\text{offpath}}}{C_{\text{onpath}}}$	$B = \prod b_i$
effort	$f = gh$	$F = GBH$
effort delay	f	$D_F = \sum f_i$
parasitic delay	p (see Table 4.3)	$P = \sum p_i$
delay	$d = f + p$	$D = \sum d_i = D_F + P$

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Limitation of Logical Effort

- Chicken and egg problem.
  - Need path to compute G.
  - But don’t know number of stages without G.
- Simplistic delay model.
  - Neglects input rise time effects.
- Interconnect is not taken into account.
  - Iteration required in designs with wire.
  - Best suited to high-speed regular array with little interconnect wires.
- Maximum speed only.
  - Not minimum area/power for constrained delay.

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Timing Analyzer Delay Models

- Slope-based linear model:
  - Linear delay model but include effect of input signal slope.
  - Suffers limitations of linear models, therefore not generally used.
- Nonlinear Delay Model:
  - Table storing delay vs load capacitances, input slope etc.
  - Use table lookup and interpolation.
  - Cannot handle complex RC network or noise events.
- Current Source Model:
  - Models output current as non-linear current source.
  - Liberty Composite Current Source model (CCSM) (widely used) - stores output current as function of input slew rate and output capacitance.

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Further Reading

- Chapter 4 of Weste & Harris – detail explanations on Logical Method.